

Introduction to Business

Financial Markets and Systems

Module Learning Outcomes

Discuss the role of business in society, the primary functions within a business, and external forces that affect business activities

- 4.1: Explain what money is and what makes it useful
- 4.2: Explain the role of banks in the U.S. monetary system
- 4.3: Describe common ways in which businesses obtain financial capital (money) to fund operations

Money

Learning Outcomes: Money

4.1: Explain what money is and what makes it useful

- 4.1.1: Explain the three key functions of money
- 4.1.2: Discuss the advantages of using money versus barter
- 4.1.3: Discuss alternatives to traditional currency used today

Key Functions of Money

Money serves three basic functions:

1. Medium of exchange: acts as an intermediary between buyer and seller.
2. Store of value: while the value of money is affected by inflation, it is a better store of value than a pair of shoes or a crop of bananas.
3. Unit of account: it's the ruler by which other values are measured.



Barter and the Double Coincidence of Wants

Without money, how would people exchange goods and services?

Economies without money typically engage in **barter**, trading one good or service for another.

Bartering is highly inefficient and requires a **double coincidence of wants**, which means a situation where two people each want some good or service that the other person can provide. Such a trade is likely difficult to arrange.

Another problem with barter is that it doesn't allow people to easily enter future contracts to purchase goods and services.

Commodity vs. Fiat Money

Commodity Money consists of objects that have value in themselves as well as value in their use as money. Gold is an example of a material that has value in itself but has also been used as a commodity money in the past.

Fiat Money is a legal tender whose value is backed by the government that issued it. The United States' paper money – like the dollar bill – carries this statement: "This note is legal tender for all debts, public and private".

Virtual Currencies

Virtual currencies, such as Bitcoin, are using the traditional concept of "money" but as an alternative to the established Federal Reserve System.



Although gaining in popularity, these virtual currencies are unregulated and pose some serious risks to those using this medium of exchange.

Cryptocurrencies

The most credible cryptocurrencies are those based on blockchain technology that records, validates and stores transaction information across a decentralized and distributed network of personal computers.

Although there were earlier versions, the first cryptocurrency to gain broad market attention was Bitcoin, launched in 2009.[2]

Retailer acceptance of bitcoin is still limited to early adopters, including a few major retailers such as Microsoft, Overstock and Newegg.[3] However, Bitcoin's value volatility make it more appropriate as a speculative investment than for use as money.

Cryptocurrency Risk Factors

- Cryptocurrency is not backed by the faith and credit of a trusted individual or institution
- Cryptocurrency is not legal tender (has limited acceptance)
- There is no central authority with responsibility for the maintenance, security or reliability of the technology or ecosystem
- There is currently no legal framework to provide recourse for consumers impacted by fraud, theft or other security breach

Mobile Commerce and Mobile Payment Systems

Mobile payment generally refers to means of accessing financial services and conducting transactions on a mobile device. Instead of paying with cash, check, or credit cards, a consumer can use a mobile phone to pay for a wide range of services and digital or hard goods.

Examples: Apple Pay, Google Wallet

A working definition of m-commerce is sales transactions made on a smartphone or tablet. Using this definition, research firm eMarketer projects that m-commerce will represent approximately 50% of U.S. e-commerce sales in 2020, up from approximately 35% in 2017.



Regionally Based Currency

Regions develop “currencies” to support a range of local economic development and sustainability initiatives.

The Schumacher Center has been involved in the development of a number of local currencies, including BerkShares, a currency specific to Western Massachusetts. Launched in 2006 and still in operation, The New York Times referred to the BerkShares program as a “great economic experiment.”

The program currently involves a network of four community banks, approximately 400 locally owned business and local citizens. BerkShares can be obtained at participating bank branches at an exchange rate of $0.95 to one BerkShare and BerkShares can be exchanged for U.S. dollars at the same exchange rate. BerkShares can be spent at face value—that is, 100 BerkShares equals $100—with participating businesses and can be used by private parties as payment for services, if desired.

Class Discussion: A New Currency?

- What are some things that could be used as currency?
- Remember the functions of money and think about what would work well as the basis of a financial system.

Practice Question 1

Money has taken a range of forms across centuries and cultures – some more practical than others. In order to be useful (and support economic growth) money needs to serve the following functions:

A. divisible, portable, and durable

B. medium of exchange, store of value, and unit of account

C. divisible, durable, and difficult to counterfeit

D. accepted as payment for labor, goods, and services

Practice Question 2

Bartering involved trading one good or service for another rather than using money to facilitate exchange. From an economics standpoint, the greatest disadvantage to using barter is that:

A. barter emphasizes community-building rather than work

B. barter limits economic productivity and growth

C. barter promotes local versus global economic development

D. barter applies to future contracts only

Practice Question 3

Technology developments have enabled a variety of alternative currencies and payment options including cryptocurrencies, mobile wallets, brand, or regionally-based currencies. Although the underlying technologies may differ, the common denominator is:

A. They are backed by the faith and credit of the issuing government.

B. They are legal tender for all debts public and private.

C. They serve the three functions of money.

D. They are not a viable replacement for money at this time.

Role of Banks

Learning Outcomes: Role of Banks

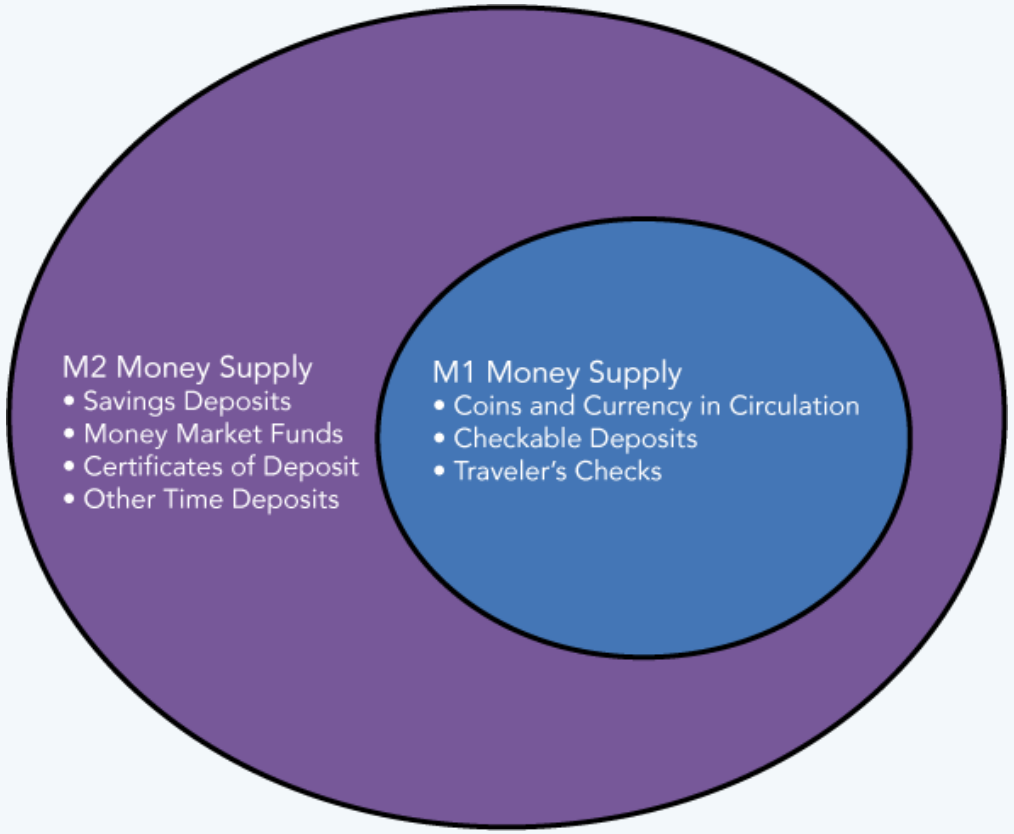
4.2: Explain the role of banks in the U.S. monetary system

- 4.2.1: Explain the difference between M1 and M2 money supply and how they are measured
- 4.2.2: Explain how banks act as intermediaries between savers and borrowers
- 4.2.3: Explain the structures and key functions of the Federal Reserve
- 4.2.4: Explain how the Federal Reserve System implements monetary policy

The Money Supply

The government uses two measures to track the money supply:

1. **M1** includes the most liquid forms of money, such as cash and checking-account funds.
2. **M2** includes everything in M1 plus near-cash items such as savings accounts, time deposits / certificate of deposits, and money market funds.



M1 Money Supply

Coins and currency in circulation: coins and bills that circulate in an economy that are not held by the U.S. Treasury, at the Federal Reserve Bank, or in bank vaults.

Demand deposits: also known as checkable deposits – the amounts held in checking accounts. There are called demand deposits because the banking institution must give the deposit holder their money "on demand" when a check is written or a debit card is used.

M2 Money Supply

Saving deposits: bank accounts on which you cannot write a check directly, but from which you can easily withdraw the money at an automatic teller machine or a bank

Money Market Funds: deposits of many individual investors are pooled together and invested in a safe way, such as in short-term government bonds

Certificates of Deposit / Time Deposits: accounts where the depositor has committed to leaving in the bank for a certain period of time, ranging from a few months to a few years, in exchange for a higher interest rate

Measuring and Tracking the Money Supply

The Federal Reserve System is responsible for tracking the amounts of M1 and M2 and prepares a weekly release of information about the money supply.

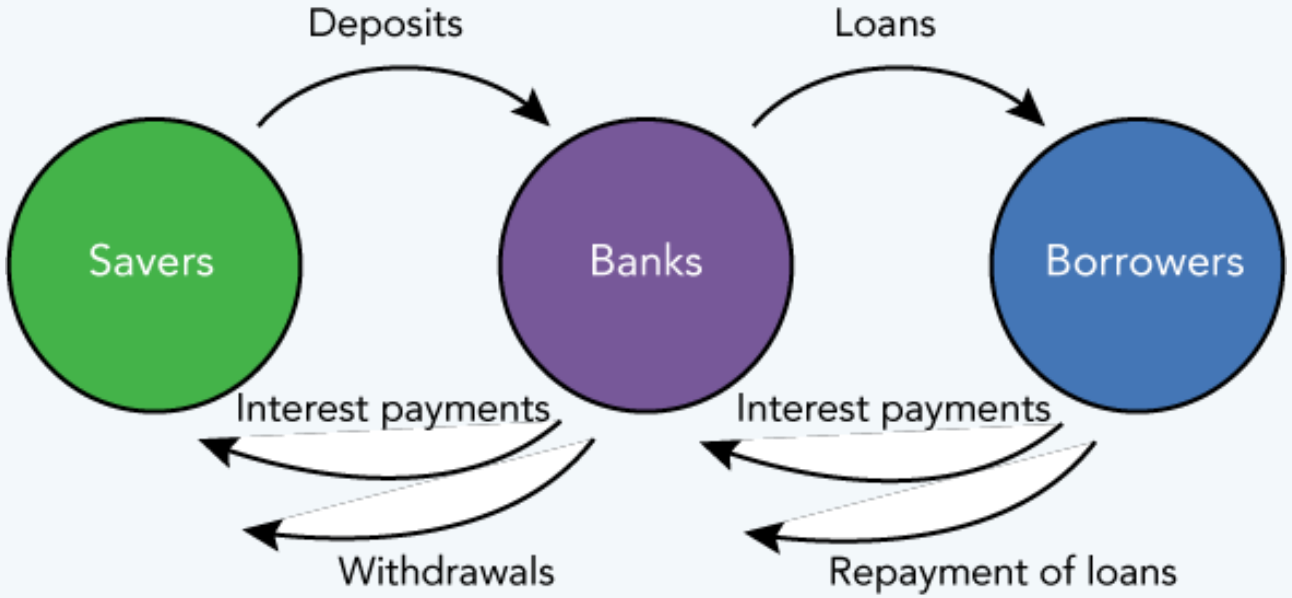
The lines separating M1 and M2 can become a little blurry. Sometimes elements of M1 are not treated alike; for example, some businesses will not accept personal checks for large amounts but will accept traveler's checks or cash.

"Plastic money" like **debit cards** act like checks, transferring money directly from the users account to seller. **Credit cards** act more like a short-term loan from the credit company to you.

Counting and tracking the money in a modern economy doesn't just involve paper bills and coins; instead, money is closely linked to bank accounts.

The Role of Banks

- Banks serve as **financial intermediaries** between savers and borrowers and direct the flow of funds between the two groups.
- All funds deposited in a bank are mingled in one big pool, which is then loaned out.
- Banks offer range of services, including checking and savings accounts, ATM services, and credit and debit cards.
- Banks sell securities and provide financial advice.



The Federal Reserve

The organization responsible for conducting monetary policy and ensuring that a nation's financial system operates smoothly is called the **central bank**. The central bank in the U.S. is called the Federal Reserve – often abbreviated as "The Fed".

Most banks are members of the Federal Reserve System. The Fed's goals include price stability, sustainable economic growth, and full employment.



Structure and Organization of the Federal Reserve

Unlike most central banks, the Federal Reserve is semi-decentralized, mixing government appointees with representation from private-sector banks.

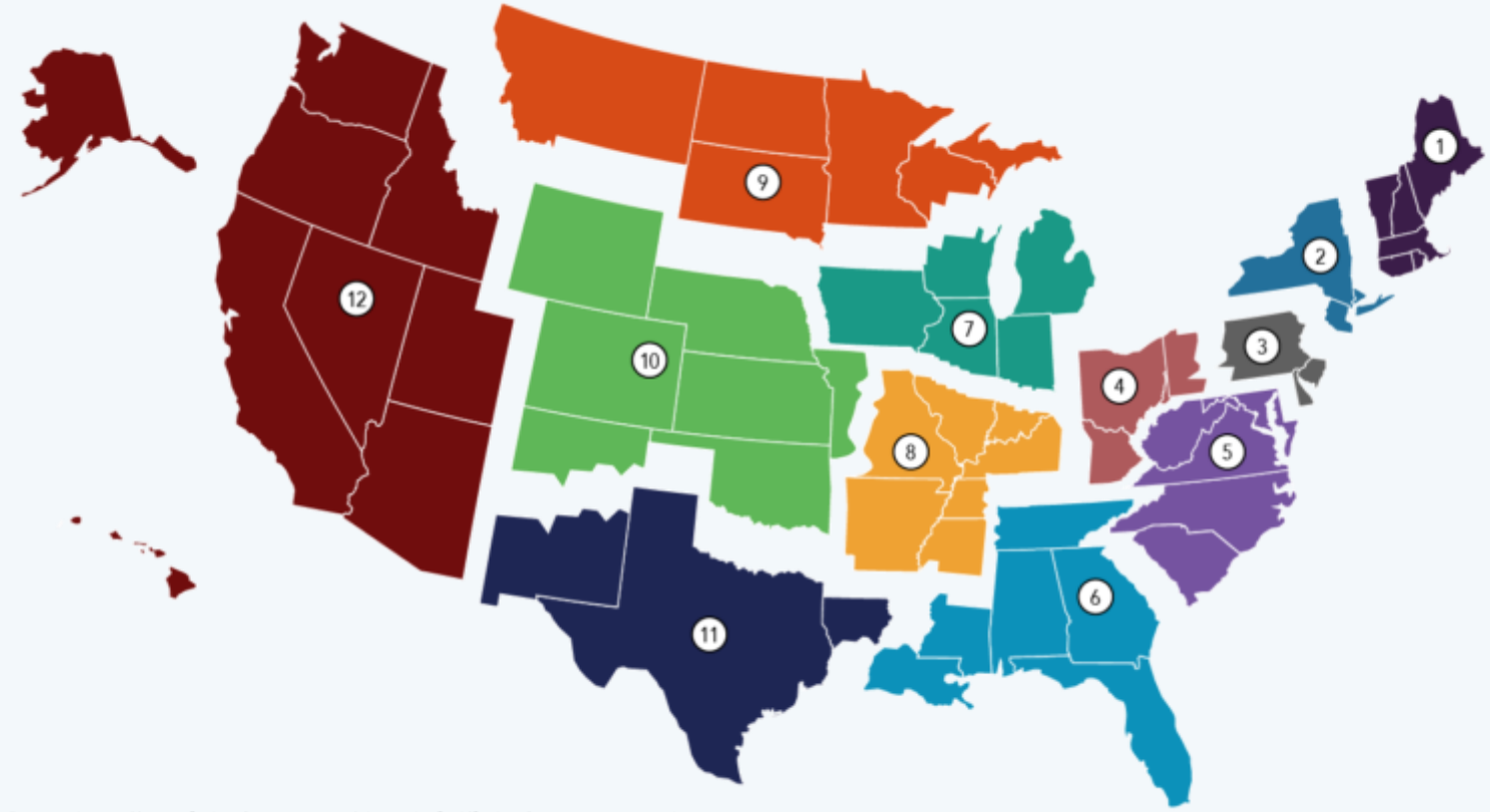
At the national level, it is run by a board of governors, consisting of seven members appointed by the president of the United States and confirmed by the Senate.

Appointments are for fourteen-year terms and they are arranged so that one term expires January 31 of every even-numbered year. The purpose of the long and staggered terms is to insulate the board of governors as much as possible from political pressure so that policy decisions can be made based only on their economic merits.

One member of the Board of Governors is designated as the Chair. The Chair of the Federal Reserve has the most immediate economic power in the world.

Federal Reserve Banks

The Fed includes twelve regional **Federal Reserve Banks** each responsible for supporting the commercial banks and economy generally in its district.



Source: https://www.federalreserve.gov/aboutthefed/federal-reserve-system.htm

The Federal Reserve Districts and Their Territories

District Number	Head Office Location	Territories Covered
1	Boston, Massachusetts	The state of Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont; and all but Fairfield County in Connecticut.
2	New York, New York	The state of New York; Fairfield County in Connecticut; and 12 counties in northern New Jersey, and serves the Commonwealth of Puerto Rico and the U.S. Virgin Islands.
3	Philadelphia, Pennsylvania	The state of Delaware; nine counties in southern New Jersey; and 48 counties in the eastern two-thirds of Pennsylvania.
4	Cleveland, Ohio	The state of Ohio; 56 counties in eastern Kentucky; 19 counties in western Pennsylvania; and 6 counties in northern West Virginia.
5	Richmond, Virginia	The states of Maryland, Virginia, North Carolina, and South Carolina; 49 counties constituting most of West Virginia; and the District of Columbia.
6	Atlanta, Georgia	The states of Alabama, Florida, and Georgia; 74 counties in the eastern two-thirds of Tennessee; 38 parishes of southern Louisiana; and 43 counties of southern Mississippi.
7	Chicago, Illinois	The state of Iowa; 68 counties of northern Indiana; 50 counties of northern Illinois; 68 counties of southern Michigan; and 46 counties of southern Wisconsin.

The Federal Reserve Districts and Their Territories (cont.)

District Number	Head Office Location	Territories Covered
8	St. Louis, Missouri	The states of Minnesota, Montana, North Dakota, and South Dakota; the Upper Peninsula of Michigan; and 26 counties in northern Wisconsin.
9	Minneapolis, Minnesota	The states of Minnesota, Montana, North Dakota, and South Dakota; the Upper Peninsula of Michigan; and 26 counties in northern Wisconsin.
10	Kansas City, Missouri	The states of Colorado, Kansas, Nebraska, Oklahoma, and Wyoming; 43 counties in western Missouri; and 14 counties in northern New Mexico.
11	Dallas, Texas	The state of Texas; 26 parishes in northern Louisiana; and 18 counties in southern New Mexico
12	San Francisco, California	The states of Alaska, Arizona, California, Hawaii, Idaho, Nevada, Oregon, Utah, and Washington, and serves American Samoa, Guam, and the Commonwealth of the Northern Mariana Islands

What Does a Central Bank Do?

The Federal Reserve, like most central banks, is designed to perform the following three important functions:

1. To conduct monetary policy
2. To promote stability of the financial system
3. To provide banking services to commercial banks and other depository institutions, and to provide banking services to the federal government

The Federal Reserve provides many of the same services to banks that banks do to their customers.

- All commercial banks have an account at the Fed where they deposit reserves
- They can obtain loans from the Fed

The Federal Reserve ensures that enough currency and coins are circulating through financial systems to meet public demand and is responsible for assuring that banks are in compliance with a wide variety of consumer protection laws.

Monetary Policy

The Federal Reserve uses *monetary policy* to regulate the money supply and the level of interest rates.

A central bank has the following three traditional tools to implement monetary policy in the economy:

1. **Open market operations**
2. **Changing reserve requirements**
3. **Changing the discount rate**

3 Monetary Policy Tools

Open market operations: take place when the central bank sells or buys U.S. Treasury bonds in order to influence the quantity of bank reserves and the level of interest rates. The specific interest rate targeted in open market operations is the *federal funds rate*. The **Federal Open Market Committee (FOMC)** makes the decisions regarding these open market operations.

Changing reserve requirements: when the central bank raises or lowers the **reserve requirement**, which is the percentage of each bank's deposits that is legally required to hold either as cash or in their vault or on deposit with the central bank. If banks are required to hold more in their reserves, they have less money to lend out.

Changing the discount rate: if the central bank raises the discount rate, then commercial banks will reduce their borrowing of reserves from the Fed, and instead call in loans to replace those reserves. Since fewer loans are available, the money supply falls and market interest rates rise.

Practice Question 4

The Federal Reserve reports on money stock measures at an individual component and summary level, distinguishing between M1 and M2 money supply categories. Which of the following statements accurately describes the primary components of the two money supplies?

A. M1 includes coins and currency in circulation and checking accounts; M2 includes M1, savings and money market accounts.*

B. M1 includes coins and currency in circulation and any accounts held at a bank; M2 includes M1 and investment account balances.

C. M1 includes coins and currency in circulation; M2 includes checking, savings and money market accounts.

D. M1 includes coins and currency in circulation and M2; M2 includes checking, savings and money market accounts.

Practice Question 5

The Federal Reserve System is the central bank of the United States, and institution established by the Federal Reserve Act of 1913, the parameters of which can be amended by Congress. As a central bank, the key functions of the Fed are to:

A. To conduct monetary policy, support financial system stability and provide banking services to banks and the federal government.

B. To conduct fiscal and monetary policy.

C. To advise the President and Congress on economic policy.

D. To managing and monitoring the money supply.

Practice Question 6

The Fed's monetary policy objectives are price stability, maximum employment and long-term economic growth. Which of the following are tools the Fed uses to implement monetary policy?

A. bank and economic regulatory authority

B. (Human) Resource action veto authority

C. proposing interest rate and reserve rate legislation

D. price level regulatory authority

Class Discussion: The Fed—More than Money

In addition to Monetary Policy and being the "bank of banks, the Fed provides excellent financial information to the public. Review the table of data provided by the Fed.

- How much has median income grown from the first year of the data to the last?
- In 1998, what was the impact of a college degree on the median income of a family household?
- In 1998, how does percentile income affect the percent of households that reported saving?
- What is the trend of median income of Black or African-American versus White non-Hispanic over the duration of the study?

- https://www.federalreserve.gov/econres/files/scf2016_tables_internal_nominal_historical.xlsx

Financial Markets and Business

Learning Outcomes: Financial Markets and Business

4.3: Describe common ways in which businesses obtain financial capital (money) to fund operations

- 4.3.1: Distinguish between bonds and bank loans as methods of borrowing
- 4.3.2: Distinguish between private and public companies
- 4.3.3: Define “stock”
- 4.3.4: Discuss how firms choose between sources of financial capital

Understanding Financial Markets and Business

Financial capital: economic resources.

Four main ways that businesses raise financial capital:

1. Early-stage capital
2. Profits
3. Bonds
4. Stocks

Early-Stage Capital

Early-stage capital is when a business owner uses his/her own money or seeks money from an angel investor or venture capital firm to get a business started.

Banks are often unwilling to lend money to start-up businesses because they are seen as too risky.

Venture capital firms make financial investments in new companies that are still relatively small in size but have substantial growth potential. These firms gather money from a variety of individual or institutional investors. All early-stage investors know that the majority of small start-ups will never hit it big, but getting in on the ground floor of a few huge successes like Netflix or Amazon can make up for those failures, which is why these investors are willing to take large risks.

Profits

Profits: If a firm's revenues are greater than costs and they are earning profits, profits from the business are reinvested in equipment, structures, or research and development.

Bonds

A **bond** is a financial contract: the borrower agrees to repay the amount that was borrowed and also a rate of interest over a period of time in the future.

Bonds are a way to raise capital through borrowing and are:

- used by corporations and governments.
- an investment for the bondholder that creates return through regular, fixed, or floating interest payments on the debt and the repayment of principal at maturity.
- traded on bond exchanges through brokers.

A **corporate bond** is issued by firms, but bonds are also issued by various levels of government.

Anyone who owns a bond and receives the interest payments is called a **bondholder.**

Stocks

Stocks are a way to raise capital by selling ownership or equity; an investment for shareholders that creates return through the distribution of corporate profits as dividends or through gains in corporate value; traded on stock exchanges through member brokers.



Corporate Stocks and Public Companies

When a company is owned by a large number of shareholders, three important questions emerge:

1. How and when does the company get money from the sale of its stock?
2. What rate of return does the company promise to pay when it sells stock?
3. Who makes decisions in a company owned by a large number of shareholders?

Initial public offering (IPO) is a firm's first sale of stock to the public.

- IPO and any stock issued thereafter provides funds to repay the early-stage investors
- IPO provides the established company with financial capital for a substantial expansion of its operations

More on Corporate Stocks and Public Companies

When a firm decides to issue new stock, it must recognize that the investors will expect to receive a rate of return. The firm can make a direct payment to its shareholders called a **dividend.**

Alternatively, a financial investor might buy a share of stock and then later sell it for a higher price – the increase in value of the stock between when it is bought and when it is sold is called **capital gain.**

A **private company** is owned by the people who run it on a day-to-day basis.

When a firm decides to sell stock, which in turn can be bought and sold by financial investors, it is called a **public company.** Shareholders own the public company.

Ownership and Public Companies

Since the owners of public companies, shareholders, are a broad group consisting of thousands or millions of investors, the shareholders vote for a board of directors, who in turn hire top executives to run the firm on a day-to-day basis.

The more shares of stock a shareholder owns, the more votes that shareholder is entitled to cast for the company's board of directors.

In theory, the board of directors helps to ensure that the firm is run in the interests of the true owners—the shareholders. However, the top executives who run the firm have a strong voice in choosing the candidates who will be on their board of directors.

Quick Review

- What is money? What makes it useful?
- What is the role of banks in the U.S. monetary system?
- What are common ways in which businesses obtain financial capital (money) to fund operations?